# Review of Stat 6545 Project Proposal by Dejene Chala, Tyler Glidden, Thao Nguyen and Tobechukwu Abor

Your proposal looks pretty good. Just be sure to implement the following suggestions and comments. There may be further suggestions and comments once I review your project write-up that is due in December.

- Be sure to do all of the analyses we consider in class; even if I didn't notice they were missing from your proposal. You want to avoid having to do the missing analyses in the short period of time between when your write-up is reviewed and when you give your presentation.
- For example, be sure to do factor analysis as well as PCA
  - Let me know right away if there appears to be an analysis you simply cannot perform, for example profile analysis may not be possible
- With 50,000 observations you are likely to reject every null hypotheses
  - Still perform the analyses we learned in class
  - You will need to also consider graphical judgements and/or multivariate effect size metrics for each analysis
  - For each technique briefly discuss the common effect size metric(s) used
  - Discuss the practical importance of the effect size when it is considered
    - MANOVA; For example Wilks-based or Pillai-based $\eta^2$, and estimated parameter values
    - Multivariate Regression; For example Wilks-based $R^2$ or Pillai-based $R^2$, and estimated parameter values
    - PCA; For example  Generalized-variance index and others?
    - Factor Analysis; skip if you like
    - CCA; For example Squared canonical correlation and others?
    - Logistic regression; Bias analysis and E-values, and estimated parameter values
- With 50,000 observations clustering will be very time consuming. You will probably have to cluster based on a random sample of the data and/or other methods for clusering with very large data sets
  - Consider k-means clustering. Is it feasible for the entire data set?
- Near-linear relationships do not cause trouble for PCA, CA and other MV methods, in general. They will cause trouble for methods that depend upon $S^{-1}$, the inverse of the sample covariance matrix

  - You will need to drop some variables which are most nearly linearly related with the others. You have many variables and they are economic variables.

- The groups being compared with MANOVA need to only be externally defined groups. Do not use groups that are found by clustering.
- Logistic regression should use gender as the response
- Consider standardization for PCA
- Consider transformations to make variables more normal before checking for multivariate normality
- Do A two-way MANOVA with all of the other continuous variables as the Y variables. Here you can take Gender and Card_Type to be your factors. Be sure to use Bonferroni intervals to explore significant results when appropriate. Don't form categorical divisons of continuous variables.
- In the section titled "Tree-Based Regression and Predictive Comparison, Logistic regression" you need to consider extensions of multivariate multiple linear regression **not** multiple linear regression. Look for and use multivariate extensions of XGBoost, Tree-Based Regression and Random Forests. By this I mean you need to use (i) XGBoost for Multiple Outputs, (ii) Multivariate Response Regression Trees and (iii) Multivariate Random Forests.